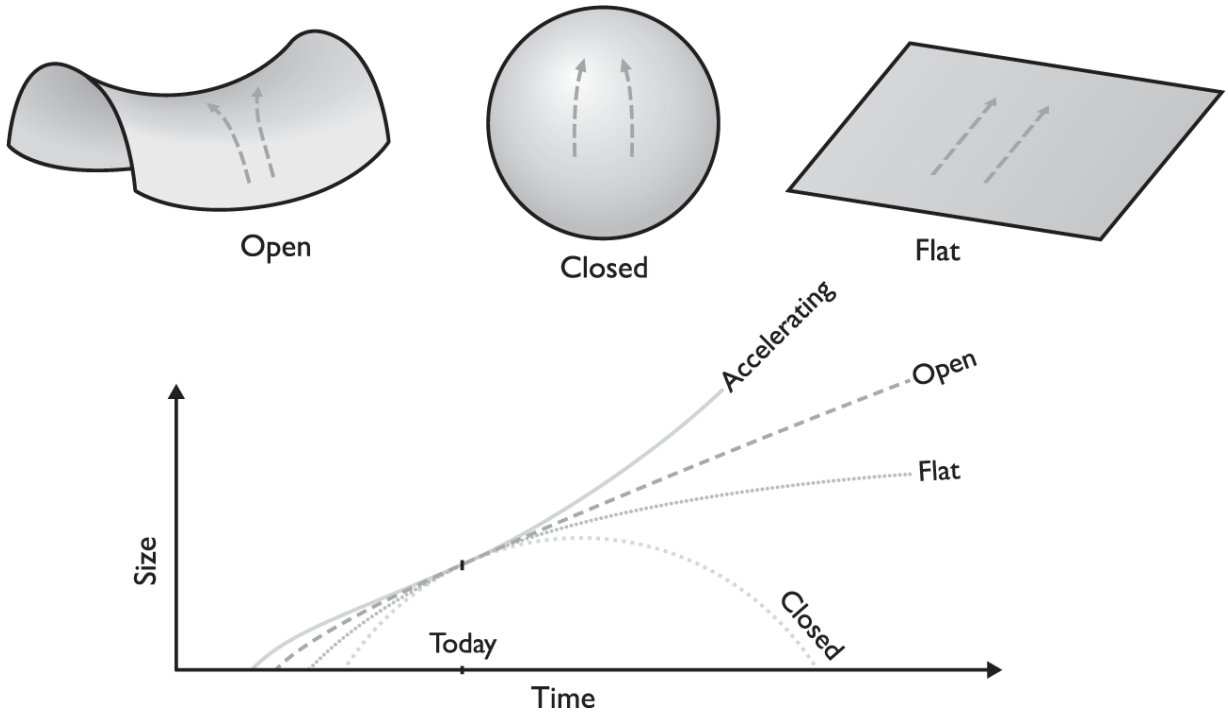


Figure 10: Open, closed, and flat universes and their evolution over time. The diagrams above indicate the shape of space for three different cosmic models. In an open universe, parallel light beams diverge over time. In a closed universe, they converge. In a flat one, they remain parallel. The different geometries correspond to different fates of the cosmos as shown in the graph. In the closed case, there is enough gravity to cause the cosmos to recollapse, whereas in the open case, the expansion wins out and the universe expands forever. A perfectly balanced flat universe continues expanding but always slowing in its expansion rate. However, if a universe contains dark energy, its expansion can accelerate (while the geometry of space remains flat).

The measurements were checked, and rechecked, but they kept forcing physicists to the same conclusion. The expansion was accelerating.

These were desperate times, and they called for desperate measures. So desperate, in fact, that the astronomers invoked the existence of a vast cosmic energy field that could make the empty vacuum of space itself have an intrinsic outward push in all directions—a previously undetected property of spacetime that would make the universe expand forever, all on its own, from an ever-present energy source, never depleting. A *cosmological constant*.

NOT-SO-EMPTY SPACE

Unlike most monumental revisions of the foundations of physics, the cosmological constant was not a new idea at all. It was, in fact, originally